

FPGA Design Using VHDL

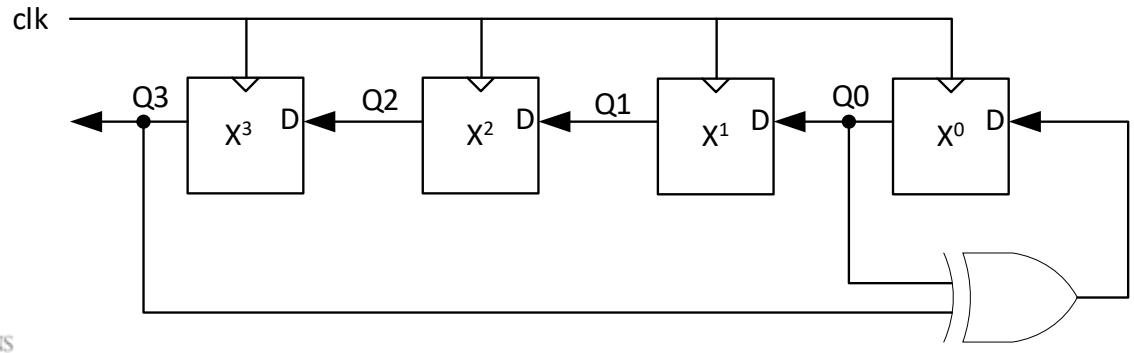
Module 8F: LFSR Design Example

LFSR Design From Polynomial (1)

- Assume 4-bit polynomial defined by $x^3 + 1$
- Assume the initial seed is "1111"
- Design the LFSR block diagram for this polynomial and determine if it is maximal length
- A few things to notice when solving this design:
  - This LFSR will need to be designed using four 1-bit flip flops
  - The flip flops are connected in series – output of one flip flop is input into next flip flop
  - The output of **some** the flip flops are combined into an XOR gate and fed back to the input of the first flip flop

LFSR Design From Polynomial (2)

- The first step is to write out the polynomial defining ALL polynomial coefficients
  - $x^3 + 1 = 1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0$
- Next step is to define the series of flip flops, direction of data flow is important
  - Direction of data flow defines how the flip flops are labelled
  - Coefficients determine which output of the flip flops are fed back into the XOR gate for the feedback



Maximal Length?

- Is this maximal length?
- Start with the seed, determine the next value
- If all 4-bit values are uniquely presented before repeating, it's maximal length
- In other words, if it repeats every 15 values, it's maximal length
- Why 15?
  - $2^4-1 = 15$

Q3	Q2	Q1	Q0	Value
1	1	1	1	15

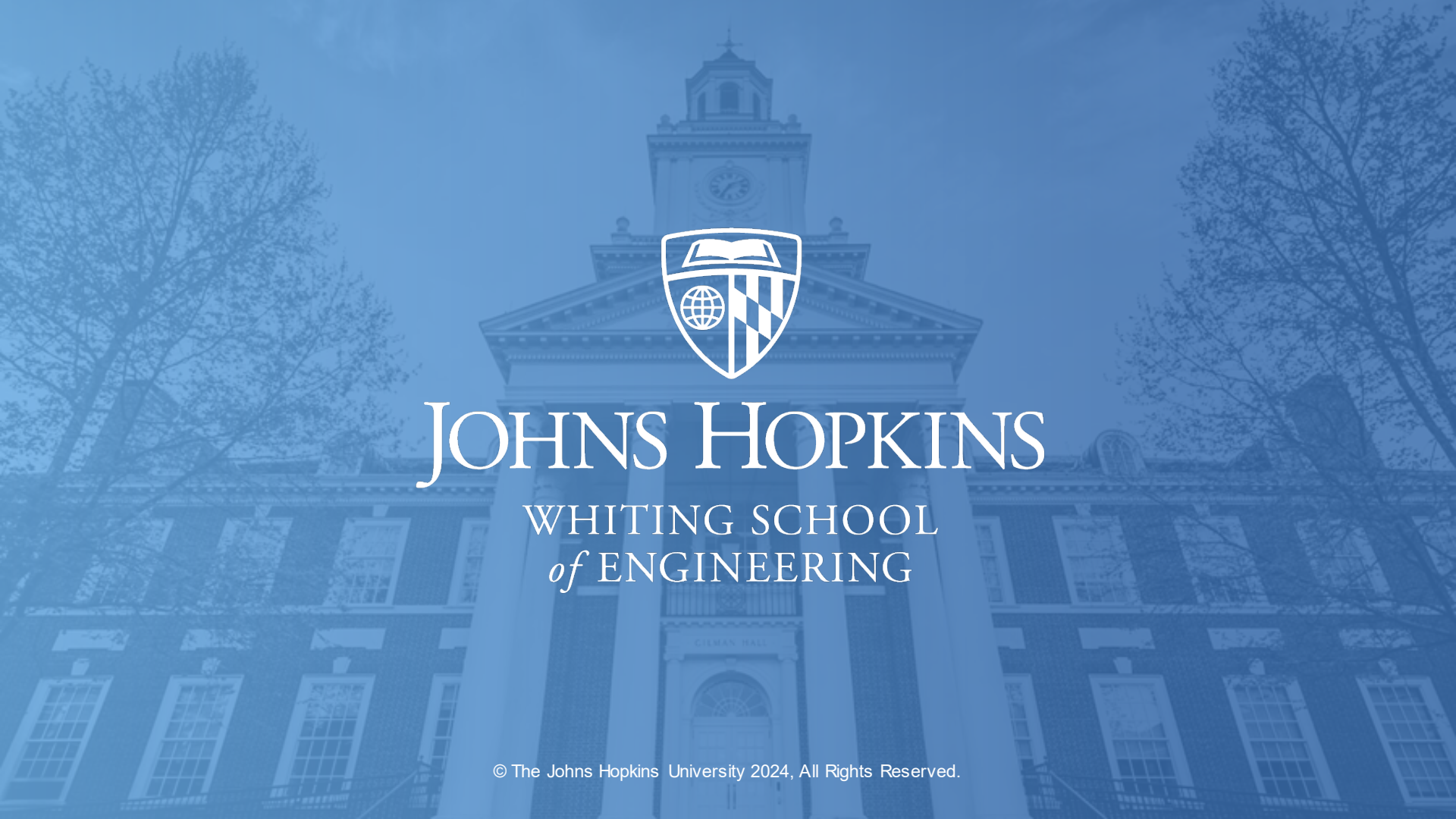
Q3	Q2	Q1	Q0	Value
1	1	1	1	15
1	1	1	0	14

Q3	Q2	Q1	Q0	Value
1	1	1	1	15
1	1	1	0	14
1	1	0	1	13

Q3	Q2	Q1	Q0	Value
1	1	1	1	15
1	1	1	0	14
1	1	0	1	13
1	0	1	0	10

Q3	Q2	Q1	Q0	Value
1	1	1	1	15
1	1	1	0	14
1	1	0	1	13
1	0	1	0	10
0	1	0	1	5
1	0	1	1	11
0	1	1	0	6
1	1	0	0	12
1	0	0	1	9
0	0	1	0	2
0	1	0	0	4
1	0	0	0	8
0	0	0	1	1
0	0	1	1	3
0	1	1	1	7
1	1	1	1	15

Repeats after 15 values



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